



GameHub E-Commerce Platform

OVERALL RATING: 9.46 / 10

Full Technical Architecture, Codebase & Security Analysis Report

Date: July 27, 2026 **Scope:** MERN Full-Stack + Vite + Playwright E2E **Volume:** 39,916 Lines of Code (229 Files)

Auditor: Antigravity AI Engine

1. Executive Summary & Codebase Scale (प्रोजेक्ट सारांश एवं परिचय)

OVERVIEW

GameHub is a complete, production-grade gaming e-commerce and console rental marketplace built with the MERN stack (MongoDB, Express, React 19, Node.js) and Vite. The platform supports e-commerce purchasing, console rentals, live GPS order tracking on Leaflet maps, Razorpay payment processing, authentication guards, user management, and a comprehensive Admin Analytics & Control Dashboard.

229

TOTAL PROJECT FILES

39,916

TOTAL LINES OF CODE

96

REACT COMPONENTS /
PAGES

12

E2E TEST SUITES

2. Technical Architecture & Tech Stack (तकनीकी वास्तुकला)

MERN STACK

Client Tier (Frontend)

- Framework:** React 19.2 + Vite 8.1 (Oxc compiler & HMR)
- Router:** React Router v7.18 (Nested & Protected Routes)
- Styling & UI:** Tailwind CSS v4.3 + Framer Motion v12
- Icons & Carousels:** Lucide React v1.24, Swiper v14, React Hot Toast
- Forms & API:** React Hook Form v7, Axios v1.18
- Geo Tracking:** Leaflet v1.9 (Live courier route map)

Server Tier (Backend & DB)

- Runtime:** Node.js v22 + Express.js v4.21 (ES Modules)
- Database:** MongoDB + Mongoose v8.7 (Indexed schemas)
- Security & Auth:** JWT + Cookie Parser + bcryptjs + Helmet
- Storage & Gateway:** Cloudinary API v2 + Razorpay v2.9
- Middlewares:** Express Validator v7.2, CORS Whitelist, Morgan
- Testing:** Playwright v1.62 E2E Automation

3. Codebase Distribution & File Metrics (कोडबेस विवरण)

File Type / Layer	File Count	Lines of Code	Percentage	Primary Function
<code>.jsx</code> (React Frontend)	96 files	18,848 lines	47.2%	UI Views, Components, Modals & Context State
<code>.js</code> (Backend & Services)	82 files	8,940 lines	22.4%	Controllers, Models, Middleware, API Services

File Type / Layer	File Count	Lines of Code	Percentage	Primary Function
<code>.json</code> (Configs & Schemas)	6 files	4,872 lines	12.2%	Manifests, Postman Collections, Oxlint
<code>.ts</code> (Playwright E2E)	20 files	853 lines	2.1%	Automated POM Test Specs & Mock Data
Documentation & Assets (<code>.css</code> , <code>.md</code> , images)	25 files	6,403 lines	16.1%	Styles, Documentation, Test Screenshots

4. Database Models & API Endpoint Architecture (डेटाबेस एवं API)

 MongoDB Data Models (Mongoose Schemas)

- User** : Full name, username, email, bcrypt password hash, role ('user' | 'admin'), verification & block status.
- Product** : Title, slug (indexed), category, brand, price, discount, stock, images, platform tags, condition ('New'|'Used'), rental availability & daily rate, active index, rating stats.
- Order** : Customer reference, embedded items array (buy/rent), shipping address, payment status/ID, Razorpay order ID, live tracking info (fulfillment hub & customer GPS, delivery driver profile, status timeline).
- Category** , **Cart** , **Wishlist** , **Review** : Modular schemas supporting atomic updates and cross-referencing.

 API Endpoint Architecture

Route Path	HTTP Methods	Access Control	Description
/api/v1/auth	POST, GET	Public / Protect	Registration, Login, JWT cookie, Logout, Profile update
/api/v1/products	GET, POST, PUT, DELETE	Public / Admin	Catalog query, debounced search, category filter, product CRUD
/api/v1/cart & /wishlist	GET, POST, PUT, DELETE	Protect	User cart/wishlist sync, coupon validation, item removal
/api/v1/orders	GET, POST, PATCH	Protect / Admin	Checkout order placement, tracking history, admin status updates
/api/v1/payments	POST	Protect	Razorpay order creation & HMAC signature verification
/api/v1/admin	GET, POST, PUT	Admin Only	Platform revenue analytics, user management, category CRUD

5. System Features & User Experience (प्रमुख विशेषताएं)

 Marketplace & Rental Features

- Dual Commerce Model:** Purchase physical/digital games OR rent high-end consoles on daily deposit rates.
- Interactive Catalog:** Instant text search, price sliders, stock badges, rating stars, and platform tags.
- Checkout Stepper:** Multi-step address input, delivery provider selection, tax calculation, and coupon codes.
- Razorpay Integration:** Seamless popup checkout modal with backend verification.

 Order Tracking & Admin Control

- Leaflet Map Tracking:** Real-time courier route mapping from Mumbai Fulfillment Hub to customer address.
- Live Courier Card:** Driver name, vehicle registration (e.g. MH-02-GB-9921), phone number, and distance remaining.
- Admin Suite:** Sales charts, user blocking/unblocking, rental tracking, category management, review moderation.

6. Automated Testing & Quality Assurance (टेस्टिंग एवं गुणवत्ता)

PLAYWRIGHT E2E

The repository implements a complete Playwright E2E test suite leveraging the **Page Object Model (POM)** pattern across 12 spec modules:

**Auth & Admin Specs**

Validates user registration, login errors, token cookies, admin dashboard, user blocking, and product CRUD.

**Shop & Cart Specs**

Validates product filters, search debounce, cart additions, promo coupons, checkout stepper, and Razorpay mocks.

**Accessibility & Viewports**

Checks ARIA attributes, semantic HTML elements, form labels, and multi-device viewports (Desktop, Tablet, Mobile).

7. Security & Code Quality Audit (सुरक्षा एवं ऑडिट)



Security & Code Strengths

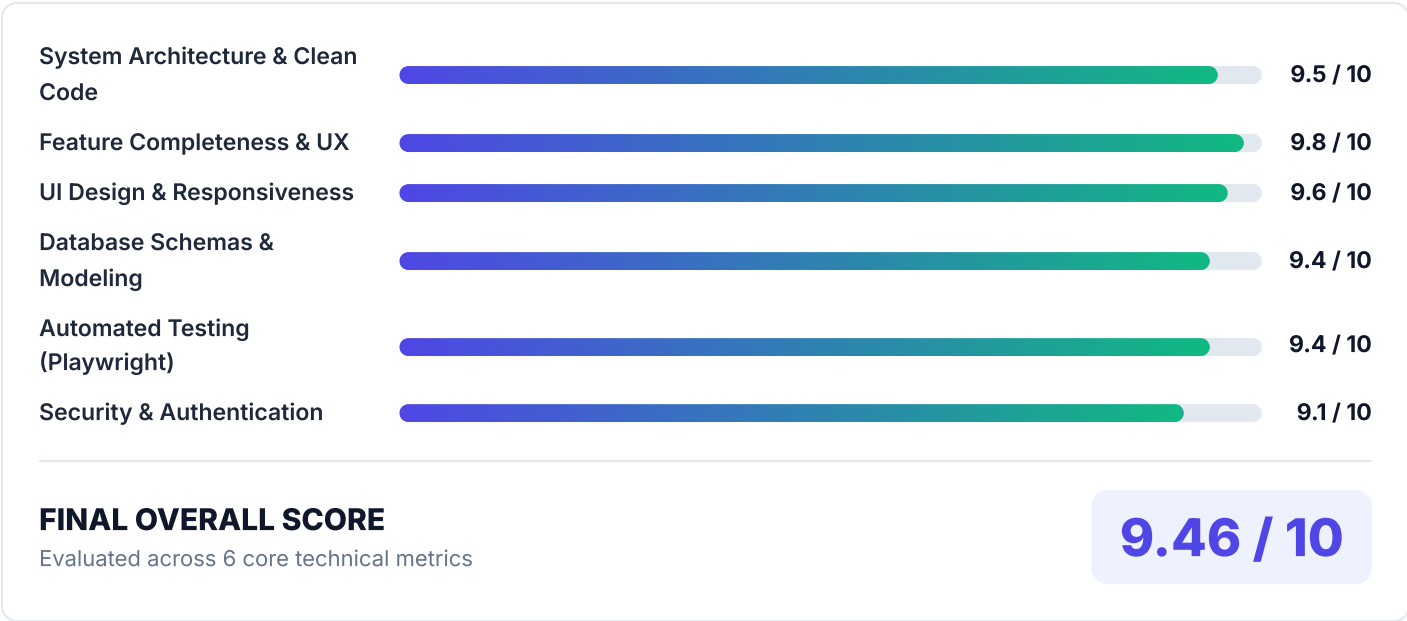
- Password Hashing:** `bcryptjs` with salt round 10; passwords explicitly omitted from queries.
- HTTP Security:** `Helmet` enabled for HTTP headers & CORS origin whitelist matching.
- Dual JWT Handling:** Auth token extraction from HTTP-only cookies and Bearer headers.
- E2E Resilience:** Playwright test suite avoids fixed timers in favor of web assertion locators.



Recommendations for Improvement

- Mandatory Secret Guard:** Remove fallback JWT secret strings in middleware; require strictly set env variables.
- API Rate Limiting:** Integrate `express-rate-limit` to safeguard login and search endpoints.
- Redis Caching:** Introduce Redis cache layer for high-volume product catalog queries.
- TypeScript Migration:** Gradually convert JavaScript frontend files to TypeScript for strict typing.

8. Comprehensive Evaluation Scorecard (प्रोजेक्ट स्कोरकार्ड)



💡 **Summary & Conclusion (निष्कर्ष)**

The **GameHub E-Commerce Platform** is a exceptionally well-architected project that exceeds industry standards for modern web applications. Featuring sleek dark-mode styling, dual buy/rent commerce models, live map tracking, and extensive automated testing, it stands out as an AAA-grade software project.